PRAKRUTHI HARISH

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SUMMARY

Biomedical Informatics graduate with 3+ years of software engineering experience and a strong foundation in clinical data analysis, machine learning, and real-world healthcare applications. Proven success in building end-to-end informatics pipelines, predictive dashboards, and decision support tools across 7+ biomedical and clinical projects. Adept at integrating APIs, managing de-identified EHR datasets, and applying ML interpretability methods for transparent healthcare AI. Passionate about developing data-driven, patient-centric solutions that improve clinical decision-making and healthcare equity.

EDUCATION

New York University, New York, NY — **July 2024 - May 2025**
Masters in Biomedical Informatics — CGPA: 4.0/4.0

Institute of Management Technology, Ghaziabad, India
Liverpool John Moores University, Liverpool, United Kingdom — **May 2021 - April 2022**
Joint Program: PG Diploma and MBA, *Distance Learning* from UpGrad — GPA: 8.5/10

Dr. Ambedkar Institute of Technology, Bangalore, India — **Aug 2015 - May 2019**
Bachelors in Medical Electronics Engineering — GPA: 8.8/10

PROFESSIONAL EXPERIENCE

Research Analyst - Decision-Making in Obstetric Care — **Dec 2024 - Present**
NYU Langone Health — New York, NY

- Led informatics-driven analysis of OB-GYN decision-making by managing and standardizing survey data from 75 clinicians across 25 labor and delivery scenarios; structured datasets for seamless integration with EHR systems to enable downstream clinical analytics.
- Developed reproducible pipelines in R to model clinical uncertainty in cesarean vs. trial-of-labor choices; insights supported the development of OB-GYN informatics toolkits and contributed to SOP alignment, internal decision support, and peer-reviewed dissemination.

Software Engineer - Splunk Support and Dev. — **Sep 2020 - Nov 2022**
Torry Harris Integration Solutions Pvt. Ltd. — Bangalore, India

- Engineered scalable data pipelines and alerting systems for British Telecom using Splunk and ServiceNow APIs, reducing manual monitoring by 50% and accelerating issue resolution by 30%; built dynamic dashboards to support real-time decision-making across 500+ daily service calls.
- Implemented predictive analytics models (StateSpaceForecast in Splunk MLTK) to forecast customer demand and optimize resource allocation; delivered modular, well-documented code in collaboration with cross-functional teams to ensure maintainability and integration with telecom operations.

Software Engineer - Integration Specialist — **Nov 2019 - Sep 2020**
Torry Harris Integration Solutions Pvt. Ltd. — Bangalore, India

- Delivered robust system integrations using WSO2 and TIBCO across two enterprise-scale projects, streamlining data exchange between legacy and cloud systems for 5,000+ users; contributed to a 30% drop in manual data handling through reusable middleware and well-tested APIs.
- Collaborated closely with Dev and QA teams to deploy and document integration workflows, ensuring scalability and long-term maintainability of release processes in fast-paced, cross-platform environments.

PROJECTS

NYU Langone Health - Capstone Project — **New York, NY**
Estimating Optimum Continuous Glucose Monitor (CGM) Wear Time for Non-conventional Populations — **Nov 2024 - Present**

- Led a clinical informatics study at NYU Langone Health to evaluate optimal CGM wear time for non–insulin-treated Type 2 diabetes patients by analyzing de-identified data from 116 participants; developed Python pipelines to compute Time in Range (TIR) metrics and assess glycemic variability against clinical CGM standards.

- Generated real-world evidence suggesting 6–9 days of CGM wear provides reliable TIR estimates, challenging the conventional 14-day benchmark and supporting more patient-centric, cost-effective monitoring protocols in underrepresented populations.
- Earned top recognition for research impact, including finalist placement among 145 participants at NYU's Three Minute Thesis (3MT) competition and poster nomination in the top 6 at the NYC Fleischer Diabetes Symposium out of 60+ submissions.

Healthcare Data Management **NYU Course**
Analysis of Warfarin Use, INR Monitoring, and Adverse Outcomes Using SQL Impala on De-Identified NYU Langone Health Data **Oct 2024 - Dec 2024**

- Analyzed warfarin prescribing patterns, INR monitoring gaps, and adverse outcomes using SQL Impala on de-identified NYU Langone Health data; identified correlations between missed tests, delayed dosing, and elevated bleeding/stroke risks to inform safer anticoagulation protocols.

Machine Learning **NYU Course**
Breast Cancer Classification Using ML Models **Sept 2024 - Dec 2024**

- Developed and evaluated ML pipelines to classify 7,900+ breast histopathology images across multiple magnifications using statistical and texture-based features (LBP, GLCM), achieving 77.1% accuracy and AUC of 0.83 with Random Forest models on the BreakHis dataset.
- Applied LIME and SHAP for model interpretability, visualizing feature contributions to enhance transparency and support clinical adoption of AI-driven diagnostic tools in breast cancer detection.

Introduction to Bioinformatics **NYU Course**
RNA-seq Data Processing and Genome Coverage Assessment **Sept 2024 - Dec 2024**

- Executed an end-to-end RNA-seq analysis pipeline on the GSE60450 dataset, including quality control, alignment to the mm10 genome (BWA, Bowtie2), and transcript quantification using Kallisto; visualized expression profiles through PCA plots and heatmaps.
- Assessed genome coverage using Lander-Waterman theory to evaluate sequencing depth and reliability, supporting robust downstream interpretation of transcriptomic data.

Introduction to R Programming **NYU Course**
Exploring Global Happiness Through: The World Happiness Report (2024) **July 2024 - Aug 2024**

- Analyzed global happiness trends across 147 countries using R to uncover correlations between GDP, social support, and life expectancy; developed interactive visualizations and statistical models to reveal regional disparities and data-driven insights from the 2024 World Happiness Report.

Introduction to Python Programming **NYU Course**
Fitbit Data Analysis: EDA & Interactive Dashboard **July 2024 - Aug 2024**

- Built an interactive Python dashboard using Dash to analyze activity and sleep trends from Fitbit data across 30 users; conducted exploratory analysis to uncover behavior-linked health patterns and correlations between daily movement and sleep quality.

Liverpool John Moores University - Capstone Project **Liverpool, United Kingdom**
A Study on the Impact of Efficient Database Management Systems on the Smooth Functioning of Healthcare Systems **April 2022 - July 2023**

- Investigated the impact of Health IT systems, specifically Database Management Systems (DBMS), on clinical efficiency and patient care quality, focusing on how optimized data infrastructure supports timely, accurate decision-making in healthcare settings.
- Proposed actionable Health IT implementation strategies to address system limitations such as poor interoperability and data latency—recommending secure, interoperable DBMS upgrades, clinician-centered training, and integrated workflows between IT teams and care providers.

Dr. Ambedkar Institute of Technology - Capstone Project **Bangalore, India**
Powered Exoskeleton for Lower Limb **April 2018 - April 2019**

- Designed and built a low-cost, sensor-integrated exoskeleton prototype to assist lower limb rehabilitation in paralysis patients, applying biomechanical principles and Arduino-based circuitry to simulate human gait and support muscle retraining.

- Engineered patient-centered features such as ergonomic padding and adjustable support straps to enhance comfort and usability; project findings were showcased at the TEQIP-III conference on Recent Trends in Electrical and Biomedical Engineering in Bangalore.

Dr. Ambedkar Institute of Technology - Mini Project **Bangalore, India**
GSM Based Heart Monitoring and Alert System **Aug 2017 - March 2018**

- Designed a GSM-enabled health monitoring system to track heart rate, body temperature, and SpO2 in real-time, targeting geriatric and at-risk populations; integrated an alert mechanism to notify caregivers of abnormal vitals, enhancing early intervention and home-based patient safety.
- Engineered the device using an LDR-LED circuit, Atmega 328 microcontroller, and GSM modem, creating a cost-effective prototype showcased at a national intercollegiate competition co-hosted by IETE, IEEE, and Engineers Without Borders.

PUBLICATIONS

- **Harish, Prakruthi, et al.** "*EMG Sensor-Based Exoskeleton for Knee Injury Rehabilitation.*" 2024 International Conference on Smart Systems for applications in Electrical Sciences (ICSSES). IEEE, 2024.
- **Harish, Prakruthi, et al.** "*EMG and EEG Sensor-Based Exoskeleton for Knee Injury Rehabilitation.*" International Conference on Trends in Engineering Systems and Technologies (ICTEST) 2025, IEEE Kerala Chapter. (Accepted for publication)

TECHNICAL SKILLS

- **Programming Languages:** Python, R, MATLAB, Java, SPL (Splunk Processing Language)
- **Data Management:** MySQL, PostgreSQL
- **Development & Markup:** React.js, JavaScript, Shell Scripting, UNIX/Linux CLI, XML, HTML, Perl-based Regular Expressions
- **API Integration:** RESTful API consumption; experience visualizing ServiceNow API outputs for operational dashboards
- **DevOps & Tools:** Docker, Git, AWS (beginner), TIBCO BusinessWorks, WSO2 ESB
- **Visualization Tools:** Splunk, Tableau, Power BI, Dash, Plotly, Matplotlib, Seaborn, ggplot2
- **Machine Learning & Data Science:** scikit-learn, SHAP, LIME, PCA, clustering, classification (Random Forest, SVM), time-series forecasting (Splunk MLTK), pandas, NumPy
- **Bioinformatics Pipelines:** BWA, Bowtie2, Kallisto, FastQC, RNA-seq preprocessing and analysis
- **Healthcare Informatics:** Clinical Decision Support (CDS) design, alert optimization, screening logic development
- **Project Management:** Agile methodology, Microsoft Office Suite
- **Soft Skills:** Collaborative research, data storytelling, critical thinking, project coordination

ORGANIZATIONS

- Secretary of Student Council - NYU Vilcek Institute **Aug 2024 - Present**
- Member of IEEE Student Board **Aug 2016 - May 2019**